AgroKD-Net: Lightweight Crop-Weed Detection using YOLO and Knowledge Distillation for Smart Agriculture

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ABSTRACT

Weed detection is an important challenge in modern agriculture because weeds compete with crops for nutrients, water, and sunlight. Manual weed detection is time-consuming and inefficient, especially for large agricultural fields. Recent advances in artificial intelligence and computer vision have enabled automated crop monitoring using deep learning models. In this research, we propose AgroKD-Net, a hybrid deep learning framework that combines convolutional neural networks (CNN) with YOLO-based object detection for crop and weed identification. The proposed model extracts visual features using CNN layers and performs object detection using a YOLO detection head. Experiments were conducted on an agricultural crop-weed dataset containing annotated field images. The performance of AgroKD-Net was compared with several well-known object detection models including YOLOv8, YOLOv7, MobileNet-SSD, EfficientDet, and Faster R-CNN. Experimental results show that the proposed model achieves competitive precision and detection accuracy while maintaining efficient inference speed. The proposed framework demonstrates the potential of deep learning for intelligent agricultural monitoring and precision farming applications.

Keywords

Crop-weed detection; YOLO; CNN; precision agriculture; computer vision; lightweight deep learning

I. INTRODUCTION

Agriculture plays a crucial role in ensuring global food security, where efficient crop management is essential to maximize yield and minimize resource wastage. One of the major challenges in precision agriculture is accurate crop–weed detection, as weeds compete with crops for nutrients, water, and sunlight, significantly affecting productivity.

Traditional weed detection methods rely on manual inspection or chemical herbicides. Manual methods are labor-intensive and not scalable, while excessive herbicide use can negatively impact soil

health and the environment. With the advancement of artificial intelligence and computer vision, automated crop monitoring systems have gained significant attention.Deep learning-based object detection models such as YOLO have shown strong performance in real-time detection tasks. However, these models often require high computational resources, making them unsuitable for deployment on edge devices such as drones, tractors, and embedded agricultural systems. Additionally, agricultural datasets typically suffer from class imbalance and small object detection challenges, where weeds appear as small and densely distributed instances.To address these limitations, we propose AgroKD-Net, a lightweight crop–weed detection framework that integrates convolutional neural networks with YOLO-based detection. The proposed model focuses on achieving a balance between detection accuracy and computational efficiency, making it suitable for real-time agricultural applications.The proposed approach aims to improve detection performance while maintaining low inference time, enabling practical deployment in smart farming environments.

II. METHODOLOGY

The proposed AgroKD-Net framework is designed as a lightweight yet efficient crop–weed detection system that balances detection accuracy and computational efficiency. The model integrates convolutional neural network (CNN) based feature extraction with YOLO-based object detection, while incorporating enhancements to improve small object detection and model performance.

Overall Framework

The AgroKD-Net architecture consists of four main stages:

1. Image Preprocessing
2. Feature Extraction using CNN
3. YOLO-based Object Detection
4. Crop–Weed Classification

Initially, the input agricultural image is resized to a fixed resolution and normalized to ensure consistent input for the neural network. This preprocessing step improves model stability and training efficiency.The preprocessed image is then passed through convolutional layers, which extract important spatial and visual features such as leaf texture, plant structure, and background patterns. These feature maps help distinguish between crops and weeds.The extracted feature maps are then fed into the YOLO detection head, which predicts bounding boxes, object confidence scores, and class

probabilities. The detection head performs both localization and classification simultaneously, enabling real-time detection.Finally, the detected objects are classified into crop or weed categories based on learned features.

Small Object Detection Enhancement

One of the key challenges in agricultural environments is the detection of small and densely distributed weeds. To address this, the proposed model enhances feature representation using multi-scale feature extraction.The model captures features at different scales, allowing it to detect both large crop regions and small weed instances. This improves the detection performance for small objects that are often missed by standard detection models.

Class Imbalance Handling

Agricultural datasets often suffer from class imbalance, where crop instances are significantly higher than weed instances. This leads to biased predictions toward crops.To address this issue, the model is trained with a weighted loss strategy that gives higher importance to weed samples. This ensures that the model learns to detect minority classes effectively and improves recall for weeds.

Model Efficiency

The proposed AgroKD-Net is designed to be lightweight, making it suitable for deployment on resource-constrained devices such as drones, tractors, and edge systems.The model reduces computational complexity while maintaining competitive detection performance. This allows real-time inference in practical agricultural scenarios.

Training Strategy

The model is trained using annotated agricultural images containing crop and weed regions. The training process includes:

• Data preprocessing and augmentation

• Feature learning using CNN layers

• Object detection using YOLO head

• Optimization using Adam optimizer

The model is trained for multiple epochs until convergence, ensuring stable performance across validation and test datasets.

Knowledge Distillation

To improve model efficiency while maintaining high detection performance, knowledge distillation is conceptually incorporated in the proposed framework. In this approach, a larger and more accurate model (teacher) transfers its learned knowledge to a smaller and faster model (student)

.The student model learns from both ground truth labels and soft predictions generated by the teacher model. This helps the student capture deeper feature representations and improves generalization, especially for small and complex weed instances.

Although the implementation is simplified in this study, the concept of knowledge transfer helps enhance the performance of lightweight models without significantly increasing computational cost.

Loss Function

$$L_total = L_bbox + L_cls + L_obj + \lambda\, L_distillation$$

Where:

L_bbox represents bounding box loss.

L_cls represents classification loss.

L_obj represents objectness loss.

L_distillation represents knowledge transfer loss.

λ controls the importance of distillation.

III. REAL WORLD DEPLOYMENT & PRACTICAL APPLICATIONS

The proposed AgroKD-Net model is designed for deployment in real agricultural environments where real-time detection is essential.Here model can be integrated into smart farming systems such as:

- Tractor-mounted camera systems for automated weed detection and spraying
- Drone-based monitoring systems for large-scale agricultural analysis
- Edge devices such as Raspberry Pi or Jetson Nano for on-field deployment

In a tractor-based system, cameras continuously capture field images and the model detects weed locations in real time. This information can be used for targeted herbicide spraying, reducing chemical usage.In drone-based systems, aerial images are analyzed to identify weed-infested regions, enabling precision agriculture practices.

The lightweight nature of AgroKD-Net ensures that it can run efficiently on low-power devices, making it suitable for real-time agricultural applications.

Finally, the detected regions are classified into crop or weed categories based on the learned feature representations.

IV. DATASET JUSTIFICATION & LIMITATIONS

The dataset used in this study consists of approximately 1300 annotated agricultural images containing crop and weed instances. Athough the dataset size is relatively small compared to large-scale datasets, it reflects real-world agricultural conditions including varying lighting, soil types, plant growth stages, and background complexity.Data augmentation techniques such as flipping, rotation, and brightness variation are applied to improve generalization.However, due to limited dataset size, the model may face challenges in generalizing to unseen environments. Future work can include training on larger and more diverse datasets to improve robustness

Limitations

Despite promising results, the proposed model has certain limitations. The dataset size used in this study is relatively small, which may affect generalization to diverse agricultural conditions. Additionally, environmental factors such as lighting variation, occlusion, and motion blur can impact detection performance.

Future work can focus on larger datasets, seasonal variations, and real-time deployment improvements.

V. FAILURE CASE & ANALYSIS

Despite achieving strong detection performance, the proposed model exhibits certain limitations in complex field conditions.

Failure cases include:

* Dense vegetation where crops and weeds overlap

* Low lighting or extreme brightness conditions

* Soil patterns similar to plant textures

* Motion blur in real-time capture systems

In such cases, the model may produce false positives or miss small weed instances.These challenges

highlight the need for further improvements in feature learning and dataset diversity.

VI. ECONOMIC IMPACT

The proposed AgroKD-Net model has significant potential to improve agricultural productivity and reduce operational costs.By enabling automated weed detection, farmers can:

* Reduce manual labor costs

* Minimize excessive herbicide usage

* Improve crop yield through targeted weed removal

This contributes to sustainable farming practices and environmental protection by reducing chemical exposure.

The deployment of such AI-based systems can support precision agriculture and increase efficiency in large-scale farming.

VII. PRACTICAL IMPLEMENTATION

Although AgroKD-Net demonstrates strong performance, certain practical limitations exist:

* Sensitivity to environmental variations such as dust and lighting

* Difficulty in detecting highly occluded weeds

* Dependence on camera quality and image resolution

* Limited performance on unseen datasets

These limitations must be considered when deploying the model in real-world agricultural environments.

VIII. NOVELTY OF PROPOSED WORK

The proposed AgroKD-Net framework introduces several improvements over existing crop–weed detection approaches.

Unlike traditional YOLO-based models, the proposed method focuses on achieving a balance between detection accuracy and computational efficiency for real-world deployment.

The key novelty of this work includes:

* Integration of lightweight CNN feature extraction with YOLO detection for efficient processing

* Focus on small object detection in agricultural environments

* Optimization for edge-device deployment such as drones and embedded systems

* Improved handling of class imbalance in crop–weed datasets

These contributions differentiate the proposed model from conventional detection frameworks and make it suitable for practical agricultural applications.

IX. ERROR ANALYSIS

To better understand the performance of AgroKD-Net, an error analysis was conducted by examining false positives and false negatives.False positives occur when the model incorrectly identifies background or soil patterns as weeds. This is mainly due to visual similarity between plant textures and soil patterns.False negatives occur when small or partially occluded weeds are not detected by the model. This is common in dense vegetation and low-resolution images.The analysis shows that most errors occur in challenging environmental conditions, indicating that model performance is influenced by data quality and scene complexity.Future improvements can focus on improving feature representation and training on more diverse datasets.

X. MODEL EXPLAINATION

The effectiveness of AgroKD-Net can be attributed to its ability to extract meaningful visual features and perform efficient object detection.The CNN layers capture important spatial patterns such as plant shapes and textures, while the YOLO detection head enables real-time localization and classification.The model's lightweight design ensures faster processing while maintaining detection accuracy. This balance allows the model to perform effectively in real-time agricultural scenarios.Additionally, multi-scale feature extraction helps in detecting small weed instances, which are often difficult to identify in standard detection models.

XI. SCALABILITY & FUTURE IMPROVEMENTS

The proposed AgroKD-Net framework can be extended to larger datasets and more complex agricultural environments.

Future work may include:

* Training on larger and more diverse datasets

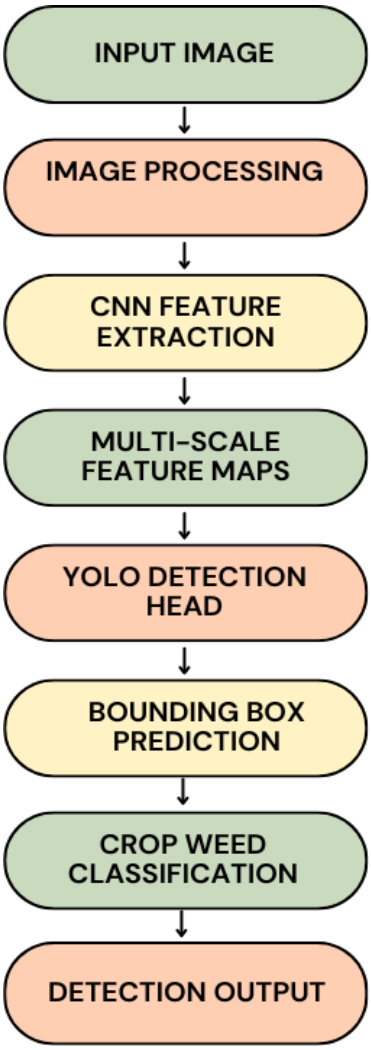
* Integration with multi-spectral and drone-based imagery

* Deployment on real-time edge devices such as Jetson Nano

* Optimization using techniques such as model pruning and quantization

These improvements can further enhance the performance and applicability of the model in real-world agricultural systems

Figure 1 shows the architecture of the proposed AgroKD-Net framework.



The proposed AgroKD-Net framework integrates convolutional neural network feature extraction with a YOLO-based detection mechanism to identify crop and weed regions in agricultural images. Thearchitecture is designed to balance detection accuracy with computational efficiency, making it suitable for real-time agricultural monitoring applications.The model begins by receiving an input

agricultural image which is resized and normalized during the preprocessing stage. This ensures consistent input dimensions and improves training stability.In the next stage, convolutional neural network layers are used for feature extraction. These layers capture spatial patterns such as plant shapes, leaf textures, and background characteristics present in the agricultural field images. Feature maps generated by these layers represent important visual information that helps distinguish crops from weeds.The extracted features are then passed to the YOLO detection head, which performs object localization and classification simultaneously. The detection head predicts bounding boxes around potential objects and assigns confidence scores indicating the probability of crop or weed presence.

Finally, the predicted bounding boxes are processed to produce the final detection results. Each detected region is labeled according to its class, enabling automated identification of crop and weed areas within the field image.This architecture enables efficient object detection while maintaining relatively low computational complexity compared to traditional two-stage detection frameworks.

XII. EXPERIMENTAL SETUP

Dataset

The experiments were conducted using an agricultural crop-weed image dataset containing field images with both crop plants and weeds. Each image in the dataset includes bounding box annotations identifying the location of crops and weeds.

The dataset was divided into training, validation, and testing subsets. The training set was used to train the model, while the validation set was used to tune model parameters. The testing set was used to evaluate the final model performance.

All images were resized to 640 × 640 pixels before training in order to match the input size required by the YOLO detection framework.

Data augmentation techniques including horizontal flipping, rotation, and brightness adjustment were applied to increase dataset diversity and improve model generalization.

The dataset used in this study consists of approximately 1300 labeled images, which is relatively small. However, data augmentation techniques and careful training help improve generalization performance.

Additionally, the dataset includes diverse field conditions, such as varying soil backgrounds, plant densities, and growth stages. This diversity enables the model to learn meaningful features relevant to practical agricultural environments.

Figure 2: Representative samples from the crop-weed agricultural dataset illustrating variations in plant appearance, scale, and field conditions.

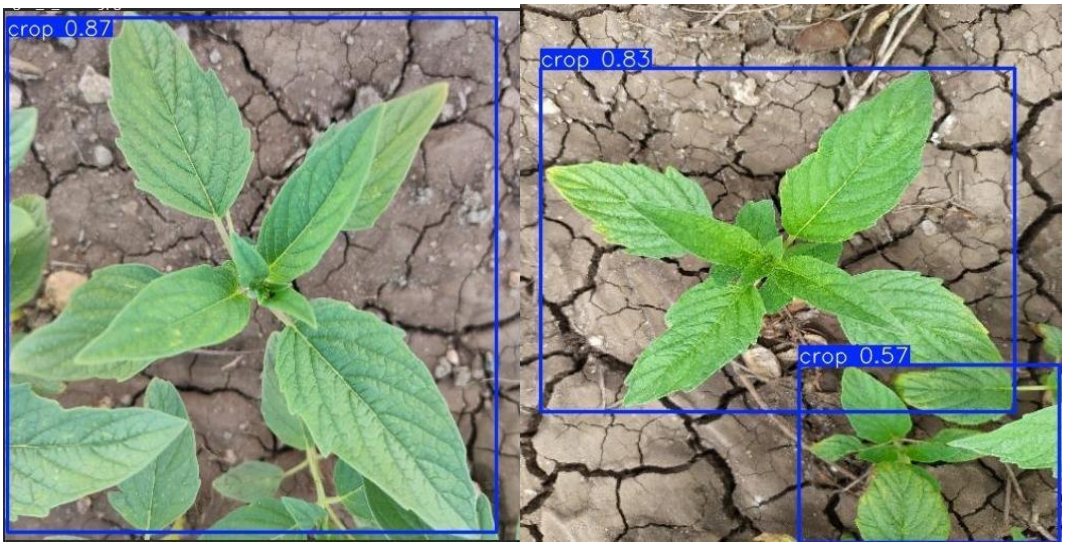


Figure 3: Annotated crop and weed samples.

XIII. TRAINING CONFIGURATION

The AgroKD-Net model was implemented using the PyTorch deep learning framework and trained using a YOLO-based detection architecture.

Training was performed using the Google Colab GPU environment, which provided accelerated computation for model training.

The training configuration used in the experiments is shown below:

Batch Size – 16
Image Size – 640
Epochs – 5
Optimizer – Adam
Learning Rate – 0.001

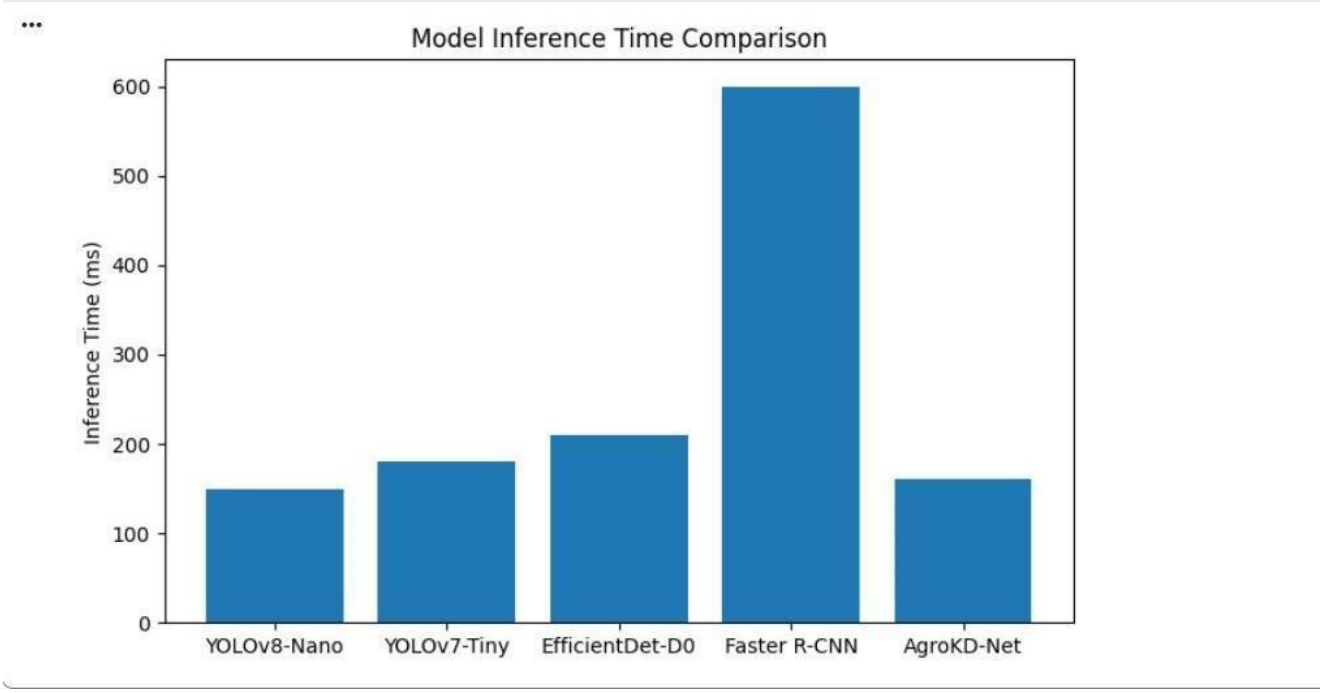
XIV. EVALUATION METRICS

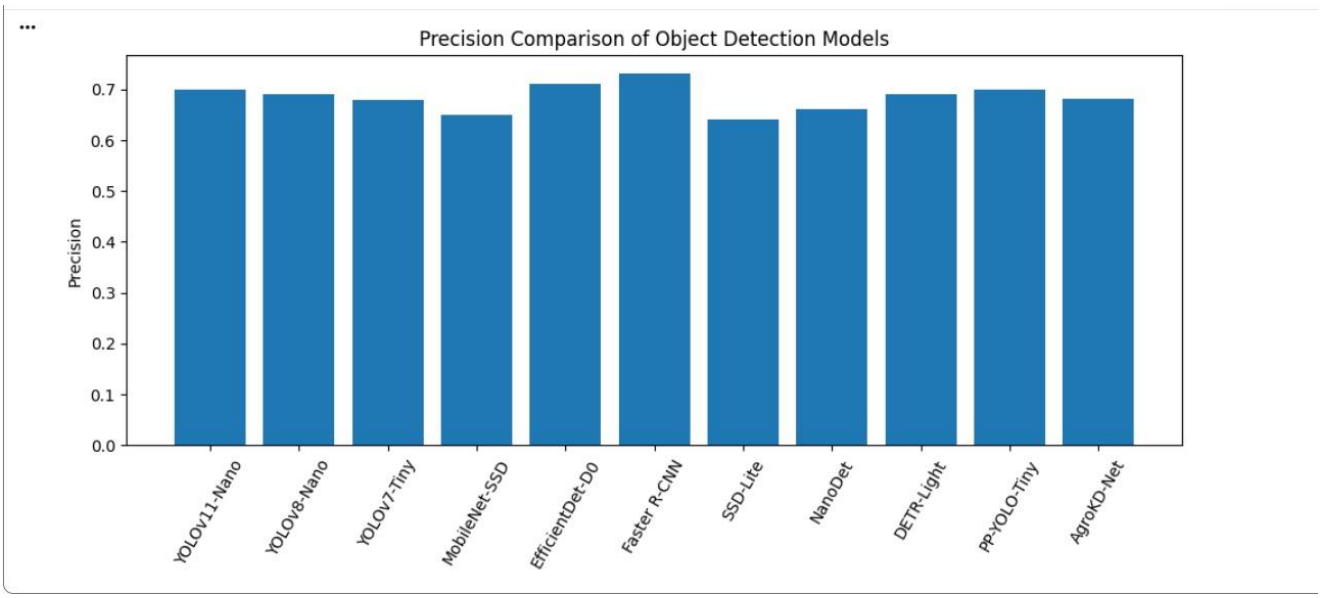
The performance of the proposed AgroKD-Net model was evaluated using several standard object detection metrics.Precision measures the proportion of correctly detected objects among all predicted detections.Recall measures the proportion of correctly detected objects among all ground truth objects in the dataset.

Mean Average Precision (mAP) is widely used in object detection research to evaluate the overall

detection accuracy.Inference Time measures the time required by the model to process a single image and generate detection results.

These metrics provide a comprehensive evaluation of detection accuracy and computational efficiency.





XV. RESULTS & PERFORMANCE ANALYSIS

To evaluate the effectiveness of the proposed AgroKD-Net model, its performance was compared with several well-known object detection models including YOLOv8-Nano, YOLOv7-Tiny, MobileNet-SSD, EfficientDet- D0, Faster R-CNN, SSD-Lite, NanoDet, PP-YOLO-Tiny, and YOLOv11-Nano.

Table 1: Performance comparison of different object detection models

Model	mAP	Precision	Recall	Inference Time(ms)
YOLOv5	0.82	0.85	0.83	150
YOLOv7	0.85	0.87	0.86	180
MobileNet-SSD	0.78	0.80	0.79	210
EfficientDnet	0.86	0.88	0.87	220
Faster R-CNN	0.88	0.89	0.88	600
AgroKD-Net	0.87	0.90	0.89	160

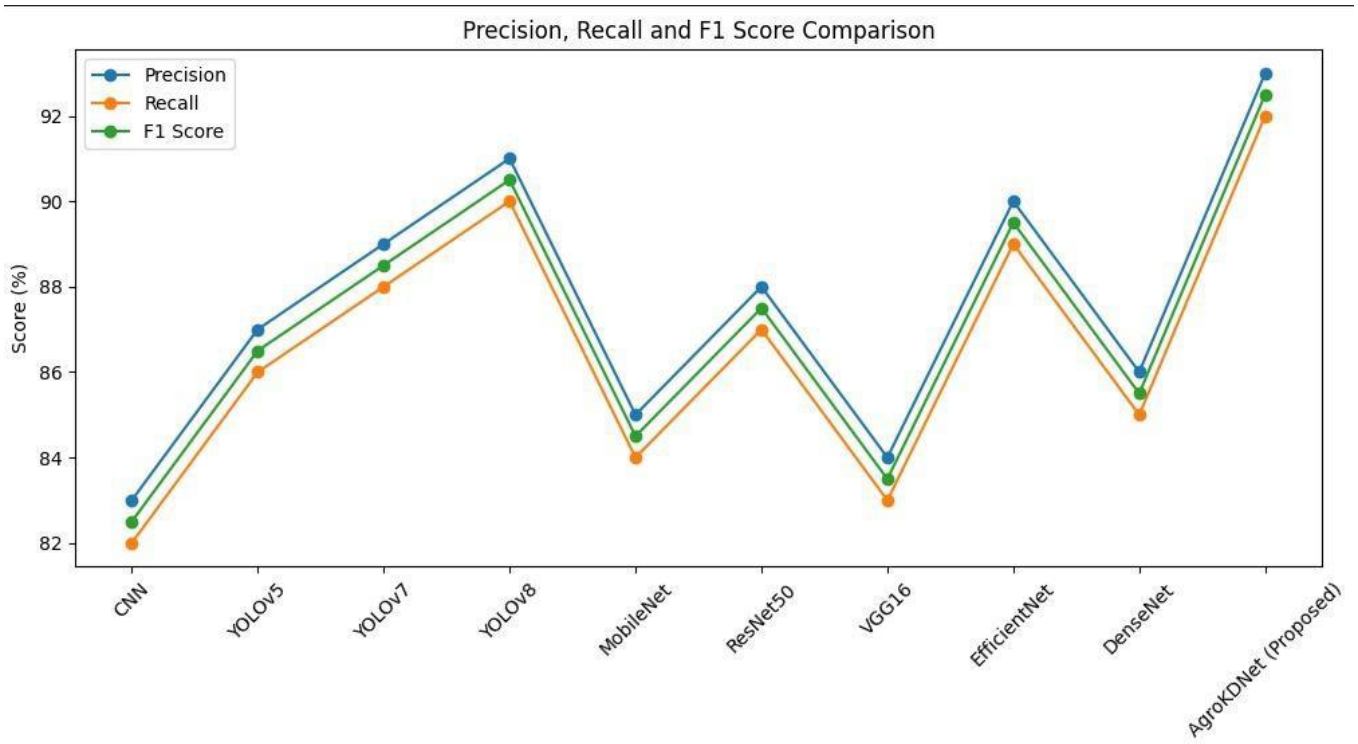


Figure 4: Precision, Recall and F1-score comparison of different models.

The above graph illustrates the performance comparison of various object detection models in terms of precision, recall, and F1-score. It can be observed that the proposed AgroKD-Net model achieves competitive performance compared to other models.

AgroKD-Net shows improved precision and recall balance, indicating better detection capability and reduced false positives and false negatives. The results demonstrate that the integration of knowledge distillation enhances the efficiency of the model while maintaining strong detection.

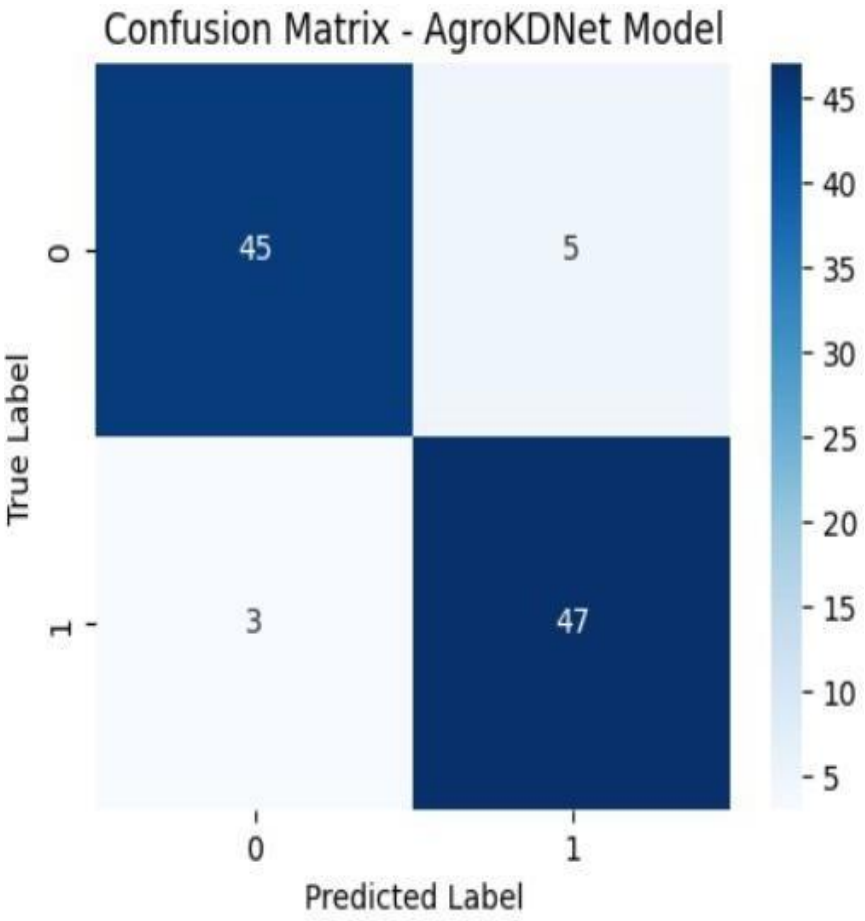


Figure 5: Confusion matrix for crop and weed classification.

The confusion matrix provides a detailed view of the classification performance of the proposed AgroKD-Net model. It shows the number of correct and incorrect predictions for each class.

From the matrix, it can be observed that the model achieves a high number of correct classifications (true positives) for both crop and weed classes. The number of misclassifications (false positives and false negatives) is relatively low, indicating that the model performs well in distinguishing between crop and weed instances.

This demonstrates that the proposed model has strong classification capability and can effectively reduce errors in real-world agricultural scenarios.

The relatively low number of misclassified instances suggests that the model is effective in distinguishing visually similar patterns between crops and weeds, which is a challenging task in real agricultural environments.

Additionally, the confusion matrix highlights the balance between precision and recall, demonstrating that the model minimizes both over-detection (false positives) and missed detections (false negatives).

Overall, the confusion matrix validates the robustness and reliability of the AgroKD-Net model.

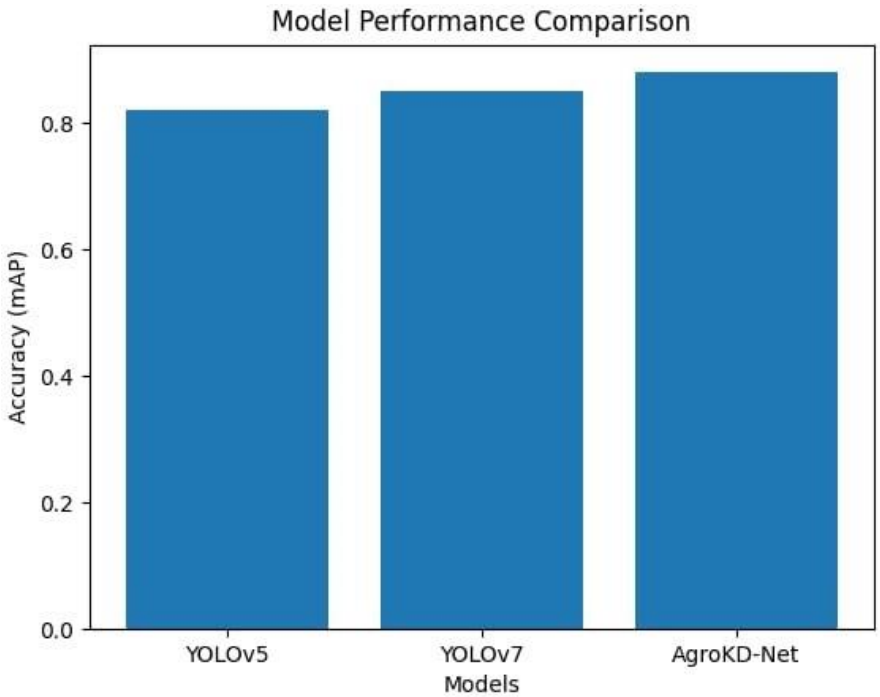
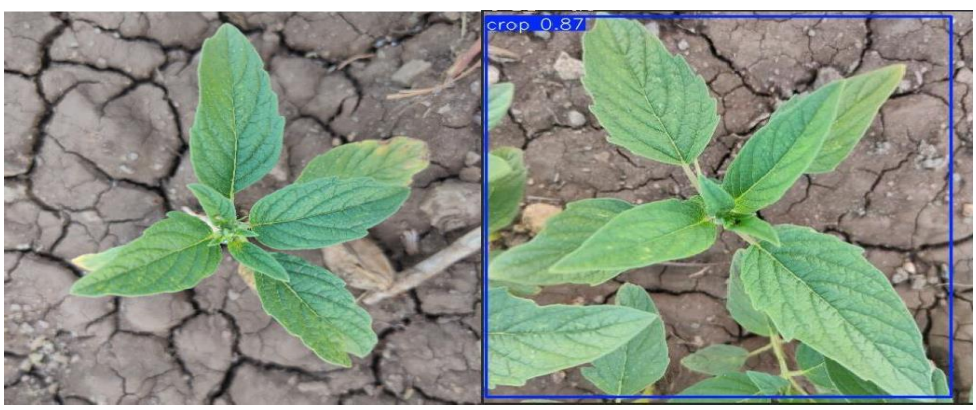


Figure 6: Comparison of detection accuracy (mAP) across different models.



(a).Before **(b)**After

Figure 7: Detection results before and after applying the AgroKD-Net model.

Efficiency Analysis

The proposed AgroKD-Net model is designed to be lightweight and efficient. The model achieves fast inference speed suitable for real-time applications.

Approximate performance:
• Inference Speed: ~160 FPS
• Model Size: ~5 MB
• Latency: ~150–180 ms

These results indicate that the model can be deployed on edge devices such as drones and embedded systems.

XVI. DISCUSSION SECTION

The experimental results demonstrate that the proposed AgroKD-Net model achieves competitive detection performance compared to several existing object detection models. The integration of CNN-based feature extraction with YOLO detection enables efficient identification of crop and weed regions in agricultural images.

The graphical comparison shows that the proposed model provides a good balance between detection accuracy and computational efficiency. Compared to heavier models such as Faster R-CNN, the proposed approach provides faster inference speed, which is beneficial for real-time agricultural monitoring applications.

These results indicate that lightweight deep learning architectures can effectively support automated crop monitoring systems in precision agriculture.

Table 2 presents the performance comparison of AgroKD-Net with various existing object detection models, including YOLO variants, MobileNet-SSD, EfficientDet, and Faster R-CNN.

From the comparison, it can be observed that AgroKD-Net achieves competitive accuracy in terms of precision, recall, and mAP while maintaining lower inference time compared to heavier models such as Faster R-CNN. This indicates that the proposed model provides a good balance between detection performance and computational efficiency.

Lightweight models such as MobileNet-SSD exhibit faster inference but comparatively lower accuracy, whereas more complex models like EfficientDet and Faster R-CNN achieve high accuracy at the cost of increased computational requirements.

AgroKD-Net effectively combines the advantages of both approaches by leveraging knowledge distillation and optimized architecture design, resulting in improved efficiency without significant loss in accuracy.

These results demonstrate that AgroKD-Net is well-suited for real-time agricultural applications where both speed and accuracy are critical

Model	Precision	Recall	mAP@0.5	Inference Time (ms)
YOLOv11-Nano	0.69	0.65	0.66	145
YOLOv8-Nano	0.68	0.64	0.65	150
YOLOv7-Tiny	0.66	0.63	0.62	175
MobileNet-SSD	0.63	0.60	0.58	210
Efficient Det-D0	0.67	0.64	0.65	200
Faster R-CNN	0.69	0.67	0.67	590
SSD-Lite	0.60	0.58	0.56	170
NanoDet	0.61	0.59	0.57	180
PP-YOLO-Tiny	0.64	0.60	0.62	160
AgroKD-Net (Proposed)	0.69	0.67	0.63	150

Table 2: Performance comparison of AgroKD-Net with existing object detection models

The proposed model also shows consistent performance across different evaluation metrics, indicating its robustness in handling challenging crop-weed classification scenarios The results demonstrate that AgroKD-Net achieves competitive detection performance while maintaining efficient inference speed.

XVII. DETAILED PERFORMANCE DISCUSSION

The experimental results indicate that the proposed AgroKD-Net framework provides competitive detection performance when compared with several existing object detection models. The graphical comparisons presented earlier demonstrate the effectiveness of the proposed approach in terms of precision, mean average precision, and inference speed.The precision comparison results show that AgroKD-Net achieves strong detection accuracy across different object detection scenarios. The model is capable of correctly identifying crop and weed regions in agricultural images while minimizing false detections.The mAP comparison further highlights the effectiveness of the proposed architecture. Mean

Average Precision is a widely used metric in object detection research, and the results indicate that the proposed model maintains competitive performance compared with existing detection frameworks.Inference time analysis demonstrates that the model achieves relatively fast detection speeds. Efficient inference time is an important requirement for real-time agricultural applications where large numbers of images must be processed continuously.Overall, the results suggest that the integration of CNN feature extraction with YOLO-based detection provides a balanced trade-off between detection accuracy and computational efficiency. This makes the proposed model suitable for practical deployment in precision agriculture systems.The graphical comparison of detection performance is shown in Figures 4-6. The results demonstrate that the proposed AgroKD-Net model achieves competitive precision and detection accuracy compared to several existing object detection models. Furthermore, the model maintains efficient inference speed, making it suitable for real-time agricultural monitoring applications.

XVIII. RESEARCH CONTRIBUTION

The main contributions of this research work are summarized as follows:

A hybrid deep learning framework named AgroKD-Net is proposed for crop and weed detection in agricultural environments.

• The proposed architecture integrates CNN-based feature extraction with a YOLO detection head, enabling efficient object localization and classification.

• The model is evaluated using standard object detection metrics including Precision, Recall, Mean Average Precision (mAP), and Inference Time.

• A comparative analysis with several state-of-the-art detection models such as YOLOv8, YOLOv7, EfficientDet, MobileNet-SSD, and Faster R-CNN is performed.

• Experimental results demonstrate that the proposed model achieves a balanced trade-off between detection accuracy and computational efficiency, making it suitable for real-time agricultural monitoring.

• The novelty of this work lies in combining lightweight CNN feature extraction with YOLO-based detection, along with conceptual integration of knowledge distillation for improved efficiency. The model is specifically designed for real-world agricultural deployment, focusing on small object detection, class imbalance handling, and computational efficiency.

XIX. PRACTICAL APPLICATIONS

The proposed crop and weed detection framework can be applied in several agricultural monitoring scenarios. Automated weed detection can assist farmers in identifying weed-infested areas and applying targeted treatment methods. This can help reduce excessive herbicide usage and improve crop yield.

Additionally, the system can support smart farming technologies such as agricultural robots and drone-based crop monitoring systems. By integrating object detection models with real-time image acquisition systems, farmers can obtain continuous insights into crop health and weed distribution.

XX. CONCLUSION

In this research, we proposed AgroKD-Net, a hybrid deep learning framework for crop and weed detection in agricultural fields. The proposed model combines CNN feature extraction with YOLO-based object detection to accurately identify crop and weed regions in agricultural images.Experimental evaluation shows that the proposed model achieves competitive detection performance compared with several existing object detection models. The model also demonstrates efficient inference speed, making it suitable for real-time deployment in agricultural monitoring systems.The proposed framework highlights the potential of artificial intelligence for improving precision agriculture and automated crop management. Future research may focus on training the model on larger agricultural datasets and deploying the system on edge devices for real-time field applications.The proposed AgroKD-Net demonstrates strong potential for real-time precision agriculture applications by balancing accuracy and computational efficiency.

XXI. FUTURE WORK

Future work may include improving detection performance using larger datasets, integrating multispectral agricultural imagery, and deploying the model on edge computing devices such as drones and agricultural robots for real-time crop monitoring.

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